Product Requirements Document

Project: Autonomous ML Research Agent | **Version:** 1.0.0 | **Phase:** MVP

1. Executive Summary

The Autonomous Machine Learning Research Agent is a web-based, AI-driven workspace designed to eliminate the manual friction of the early-stage machine learning workflow. By utilizing the Model Context Protocol (MCP), the platform allows researchers to use a conversational interface to seamlessly discover academic literature, source corresponding datasets, and autonomously trigger local data-cleaning pipelines without context switching.

2. Target Audience & Personas

- **Academic Researchers & Students:** Need to quickly find state-of-the-art papers (e.g., on specific algorithms) and source datasets for thesis work.
- **Data Scientists & ML Engineers:** Need to rapidly transition from theoretical research to practical implementation by acquiring high-quality, pre-cleaned data.

3. User Stories

Persona	User Story	Acceptance Criteria
Researcher	I want to query a topic via chat so that the AI returns a summarized list of recent arXiv papers.	The AI extracts exact titles, authors, and PDF links without hallucinating fake papers.
Data Scientist	I want the AI to automatically search Kaggle for datasets related to the papers it just found.	The AI returns real Kaggle datasets, filtering out low-usability or empty datasets.
Developer	I want the AI to run a local script to download and clean a dataset once I approve it.	The AI triggers a local Python pipeline and confirms when the CSV is saved locally.
End User	I want a clean, responsive web interface so that I don't have to interact via terminal commands.	The UI provides chat bubbles, loading states, and clickable links for PDFs.

4. Functional Requirements

- **Conversational Interface:** A React-based chat UI supporting multi-turn conversations.
- **Semantic API Orchestration:** The LLM must autonomously decide when to call the arXiv API versus the Kaggle API based on the user's natural language request.
- **Agentic Pipeline Execution:** The system must securely trigger local shell commands to execute Python data manipulation scripts (e.g., using Pandas to clean datasets).
- **Stateful Memory:** The system must remember previous messages in the session so users can refer to prior datasets contextually.

5. Success Metrics & Constraints

- **Latency:** The LLM routing and API fetching should return an initial response to the UI within 3–5 seconds.
- **Accuracy:** 100% deterministic accuracy on academic links (zero hallucinations for URLs).
- **Core Metric:** A user can go from a vague query to a downloaded, cleaned CSV file on their local machine in under 2 minutes.